

Md. Modabbir Hossain Mahin

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About Me

Computer Science and Engineering graduate and published AI researcher with five accepted papers at IEEE international conferences spanning explainable AI, federated learning, and clinical decision support. B.Sc. thesis focused on explainable clinical risk stratification under complete and partial diagnostic panels, comparing Explainable Boosting Machines with a custom hierarchical additive deep network. Experienced in mentoring and teaching, having tutored 100+ students and led 30+ active members as President of the SUB Computing Club, organizing workshops and guiding junior students in project development and research. Seeking a lecturer position to combine research supervision with undergraduate teaching in computer science and machine learning.

Education

B.Sc. in Computer Science and Engineering – State University of Bangladesh

2022 – 2026, CGPA 4.00

H.S.C – Dr. Mahbubur Rahman Mollah College (DMRC), Dhaka Board

2021, GPA 5.00

S.S.C – I.E.T. Government High School, Dhaka Board

2019, GPA 5.00

B.Sc. Thesis

“An Explainable Clinical Decision Support System for Patient Risk Stratification with Complete and Partial Diagnostic Panels: A Comparative Study of Explainable Boosting Machines and a Hierarchical Additive Deep Network”

Supervisor: Prof. Dr. Sharmin Parveen, Dept. of CSE, State University of Bangladesh

A two-phase explainable clinical decision support system combining classical ML and a custom deep learning architecture with SHAP/LIME explainability to predict patient risk levels under complete and partial diagnostic panels.

Research & Publications

- **A Deep Space Flexible Blind Convolutional Neural Network for Super Compressed JPEG Artifacts Removal.**
QPAIN 2025.
- **FedMaternal-XAI: A Privacy-Preserving Federated Learning Framework with Cross-Method Explainability for High-Risk Pregnancy Classification.**
2026 IEEE International Conference on Future Machine Learning and Data Science (FMLDS 2026), Kobe, Japan.
- **Evaluating Digital Integrity in Healthcare AI: Explainable Boosting Machines (EBM) vs. Model-Agnostic Interpretability in Multi-Class Clinical Risk Assessment.**
2026 IEEE International Conference on Future Machine Learning and Data Science (FMLDS 2026), Kobe, Japan.
- **Privacy-Preserving and Explainable Federated Learning for Heart Disease Prediction Using Differential Privacy and Explanation Consistency Analysis.**
5th IEEE International Conference on Biomedical Engineering, Computer and Information Technology for Health (BECITHCON 2026).
- **Diag-Tactics Arena: A Comparative Study of Minimax and Alpha-Beta Pruning Under Dynamic Hazard Constraints in a Diagonal-Win Board Game.**
3rd IEEE International Conference on Computing, Applications and Systems (COMPAS 2026).
- **Calibration Over Accuracy: A Statistically Validated, Uncertainty-Aware Framework for Dengue Diagnosis Using Routine CBC Data.**
3rd IEEE International Conference on Computing, Applications and Systems (COMPAS 2026).

Teaching & Leadership Experience

- **President, SUB Computing Club** 2025 – Present
Department of Computer Science and Engineering, State University of Bangladesh
 - Lead and coordinate 30+ active members in academic and technical activities
 - Organized workshops and seminars on AI, programming, and software development
 - Mentored junior students and guided them in project development and research
 - Coordinated and supported national-level programming and tech competitions
- **Tutor** 2021 – Present
 - Taught 100+ students at SSC and HSC levels, with a focus on structured problem-solving
 - Developed strong communication and mentoring skills through consistent one-on-one instruction
- **Campus Ambassador, bKash presents SUST CSE Carnival 2026** 2026
Organized by CSE Society, Shahjalal University of Science and Technology (SUST)
 - Represented the event on campus, driving outreach and student participation
 - Coordinated promotional and logistical efforts contributing to the event's success

Technical Skills

- **Programming Languages:** Python, C, C++, Java, JavaScript
- **Core CS:** Data Structures & Algorithms, OOP, Parallel Programming (OpenMP, MPI)
- **Machine Learning & Data:** NumPy, Pandas, Matplotlib, Seaborn, CNNs, XAI (SHAP, LIME), Federated Learning
- **Web Development:** HTML, CSS, JavaScript, React (learning), Firebase
- **Networking:** OSI & TCP/IP Models, TCP vs UDP, IP Addressing, DNS, HTTP/HTTPS, Ports, REST API Fundamentals
- **Tools & Platforms:** Git, Google Colab, Jupyter Notebook, VS Code, Android Studio

Selected Projects

- **BankSync — Smart Banking Made Simple:** A web-based banking system enabling users to manage accounts, perform transactions, and securely access financial information. ([Live Demo](#)) ([GitHub](#))
- **MedRoute — A Mobile Health Solution:** A mobile application connecting users with essential medical services such as ambulance support, blood banks, ICU availability, and pharmacies in real time. ([GitHub](#))
- **Sonali Fasal — Agricultural Assistance App:** An intelligent agricultural application providing crop price prediction and plant disease detection to support farmers' decision-making. ([GitHub](#))

Languages

- Bangla (Native)
- English (Fluent)

References

Professor Dr. Sharmin Parveen

Designation	Professor
Department	Computer Science and Engineering
University	State University of Bangladesh
Email	sharmin.cse@sub.edu.bd
Qualification	PhD (India); MPhil (DU); M.Sc & B.Sc in Applied Physics & Electronics (DU)

Shah Reza M Fahad Hossain

Designation	Proctor & Assistant Professor
Department	Computer Science and Engineering
University	State University of Bangladesh
Email	shahreza.cse@sub.edu.bd
Qualification	MSc. in Mobile Computing & Communications, University of Greenwich (UK); BSc. in CSE, KUET
Specialization	Cyber Security, Blockchain, Machine Learning